AtlasIT Architecture & Scalability Strategy Report

Executive Overview

AtlasIT aims to deliver an automation■first IT and DevSecOps platform focused on SMB organizations with limited or no dedicated IT staff. The platform combines Identity Lifecycle Management (JML), Compliance Digital Twin (CDT), workflow orchestration, and evidence-driven governance. This report consolidates architecture guidance designed to scale to 100k tenants without requiring major system rewrites.

Core Product Thesis

The platform focuses on reducing operational overhead by automating identity provisioning, application access management, and compliance evidence generation. The target environment includes SMB ecosystems using HR systems (e.g., BambooHR), identity providers (Google Workspace or Microsoft), and a growing SaaS stack.

Three■Plane Platform Architecture

The recommended architecture separates the system into three logical planes: Control Plane (tenant configuration and policy management), Execution Plane (workflow orchestration and connector operations), and Evidence Plane (immutable evidence ledger and compliance state). This separation ensures scalability and prevents operational coupling as tenant count grows.

Control Plane Responsibilities

Tenant management, onboarding wizard configuration, connector catalog, policy assignment, workflow templates, billing limits, and administrative console APIs.

Execution Plane Responsibilities

Runs JML workflows, manages connector integrations, handles retries, processes queues, applies policy checks, and triggers AI analysis pipelines. Execution tasks should be queue■driven and asynchronous to support high■volume workloads.

Evidence Plane Responsibilities

Stores compliance evidence envelopes, workflow step logs, policy decisions, and immutable artifacts. Evidence should be stored in object storage with a separate metadata index for fast queries.

Scaling Model

Scalability is achieved by decoupling workflow execution, evidence storage, policy evaluation, and connector runtime. These components scale independently based on workload demand rather than tenant count alone.

Cloudflare MVP Deployment Model

Initial deployment uses Cloudflare Workers for APIs and webhooks, Durable Objects for temporary orchestration state, D1 for relational metadata, R2 for immutable evidence storage, and queue services for asynchronous task execution.

Future AWS Migration Path

If infrastructure migration occurs, the recommended mapping includes API Gateway + Lambda for API ingress, Step Functions for workflow orchestration, DynamoDB for workflow state and metadata, S3 for evidence storage, and SQS/EventBridge for event routing.

AI Governance Layer

AI components should operate as bounded analysis services capable of recommending automation improvements, anomaly detection, and workflow optimizations. All AI decisions must pass policy checks and generate evidence records.

Security and Tenant Isolation

All data stores must enforce tenant■scoped partitioning. Sensitive secrets should be encrypted and stored through a dedicated secret manager, with applications referencing secrets via secure identifiers rather than storing credentials directly.

Workflow Execution Model

All automation tasks should be executed through resumable workflows. Each workflow consists of deterministic steps that can retry, pause for approval, or resume after failure while maintaining full audit lineage.

Evidence Architecture

Evidence artifacts must contain hashes, timestamps, workflow context, policy version identifiers, and action results. Metadata should be indexed separately to allow fast query while raw artifacts remain immutable.

Design Targets for Large Scale

Architecture should support 100k tenants, tens of thousands of workflows per day, and hundreds of millions of evidence records over time while maintaining strict tenant isolation and predictable execution behavior.

Strategic Outcome

By structuring the system around automation workflows, policy governance, and evidence generation, AtlasIT becomes a governance automation runtime rather than a simple IT tool. This enables strong differentiation and long■term platform scalability.

Architecture Layer Summary

Plane	Purpose	Key Components
Control Plane	Tenant configuration and policies	API services, tenant DB, workflow templates
Execution Plane	Runs automation workflows	Queue workers, connector runtime, orchestration engine
Evidence Plane	Immutable compliance ledger	Object storage, evidence index DB, snapshot engine